Jump Closure and Limit Uniformization in the Ideal Completion of the Turing Degrees

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Abstract

The Turing jump has no fixed point on the Turing degrees: $\mathbf{a} <_T \mathbf{a}'$ for every degree $\mathbf{a}$. After passing to the ideal completion, however, a natural fixed-point phenomenon appears. We study the Scott-continuous lifting $\Gamma : \mathrm{Idl}(\mathbf{D}_T) \to \mathrm{Idl}(\mathbf{D}_T)$, given by $\Gamma(I) = \downarrow \{\mathbf{a}' : \mathbf{a} \in I\}$. Starting from the computable degree, Kleene iteration reaches its first fixed point at stage ω, namely the Turing ideal of arithmetical degrees; more generally, above $\mathbf{a}$ the least fixed point is the ideal of degrees arithmetical in $\mathbf{a}$.

To pass beyond this fixed point, we introduce a limit-uniformization operator. Although the ideal of finite jumps contains every $\mathbf{0}^{(n)}$, it does not contain the uniform limit oracle $\mathbf{0}^{(\omega)} = \deg_T\big(\bigoplus_{n<\omega} 0^{(n)}\big)$. The uniformization operator adjoins this oracle only when all finite jump degrees are present. It is monotone but not Scott-continuous. Composing jump closure with one such gate yields closure ordinal $\omega \cdot 2$; gates at $\omega, 2\omega, 3\omega, \ldots$ yield closure ordinal ω^2.

We then carry the construction through every recursive ordinal at once, indexing the uniformization gates by Kleene's system $\mathcal{O}$ of ordinal notations. The resulting operator has least fixed point HYP, the ideal of hyperarithmetic degrees, and closure ordinal exactly ω_1^{CK}, the Church–Kleene ordinal; truncating the gate family at any recursive limit λ realizes λ itself as a closure ordinal. Non-uniform closure under relativized halting problems is Scott-continuous and reaches fixed ideals, while uniform coding of an entire prior hierarchy is infinitary, discontinuous, and reopens diagonalization, all the way to the Church–Kleene boundary. This gives a domain-theoretic semantics for the successor/limit distinction in transfinite Turing-jump hierarchies and links failures of Scott continuity with closure ordinals.

Keywords. Turing jump; Turing degrees; halting problem; domain theory; Scott continuity; Turing ideals; closure ordinals; hyperarithmetic hierarchy; ordinal analysis; Church–Kleene ordinal.
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1 Introduction

The Turing jump is the canonical operation for producing a stronger degree of unsolvability from a given oracle. If $A \subseteq \omega$, its jump

$$A' = \{e : \Phi_e^A(e) \downarrow\}$$

is the halting problem relative to A. Its degree depends only on the Turing degree of A, and the induced operation

$$\mathbf{a} \longmapsto \mathbf{a}'$$

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is monotone and strictly increasing:

$$\mathbf{a} <_T \mathbf{a}'.$$

Consequently, the jump has no fixed point on the Turing degrees. This strictness is the degree-theoretic form of the recurrence of the halting diagonal: once an oracle solves one level of halting problems, machines using that oracle generate a new halting problem one jump higher.

Classical computability theory responds by iteration. The finite sequence

$$0, 0', 0'', \ldots$$

organizes the finite arithmetical hierarchy. Effective transfinite iteration along recursive well-orders produces the hyperarithmetic hierarchy. At a recursive limit ordinal λ, one does not take another ordinary successor jump; instead one forms an effective join coding the preceding stages into a single oracle. The basic example is

$$0^{(\omega)} = \bigoplus_{n<\omega} 0^{(n)}.$$

Turing's ordinal logics initiated this use of oracle computation in transfinite progressions, while the later work of Kleene and Spector supplied the machinery by which iterated jumps are organized through recursive ordinal notations [13, 12, 7].

The purpose of this paper is to reinterpret the successor/limit structure as a fixed-point phenomenon in domain theory. The central observation is that the absence of fixed points for the jump depends on the type of object to which the jump is applied. An individual degree can never contain its own jump. A collection of degrees, by contrast, can be closed under taking jumps. Turing ideals closed under the jump are classical objects, and they occur naturally as the second-order parts of ω-models of arithmetical comprehension [10].

We exploit the ideal completion of the Turing degrees. Ideal completion is a standard construction in domain theory: directed lower sets are treated as completed informational states, while principal ideals embed the original poset as compact approximants [1, 8]. Every monotone self-map of a poset has a canonical Scott-continuous extension to its ideal completion. Applied to the Turing jump, this yields

$$\Gamma(I) = \downarrow \{\mathbf{a}' : \mathbf{a} \in I\}.$$

Although the original jump has no fixed degree, Γ has a least fixed point. Starting from the computable degree, the iteration is

$$\downarrow 0 \subsetneq \downarrow 0' \subsetneq \downarrow 0'' \subsetneq \cdots,$$

and its first limit is

$$\bigcup_{n<\omega} \downarrow 0^{(n)}.$$

By Post's theorem, this is precisely the ideal of arithmetical degrees. It is already closed under the jump. The closure ordinal is exactly ω.

This supplies a precise sense in which a "Turing-jump fixed point" exists:

$$\text{no degree satisfies } \mathbf{a}' = \mathbf{a}, \qquad \text{but} \qquad \text{a domain of degrees can satisfy } \Gamma(I) = I.$$

The fixed point does not solve diagonalization by producing a universal oracle. It contains, non-uniformly, a solver for every finite level of the hierarchy.

That qualification is essential. The arithmetical ideal contains each $0^{(n)}$, but it does not contain $0^{(\omega)}$. The latter packages all finite jumps into one oracle. Once this package is admitted, its own relativized halting problem

$$0^{(\omega+1)} = (0^{(\omega)})'$$

reopens the jump hierarchy.

Our second aim is to isolate this operation order-theoretically. We call it limit uniformization. Given an increasing sequence of degrees and a degree uniformly coding that sequence, the associated operator waits until an ideal contains every member of the sequence and only then adjoins the uniform code. The requirement that all stages be present has no finite witness. Correspondingly, the operator is monotone but fails Scott continuity.

The distinction can therefore be summarized as

jump closure is Scott-continuous, whereas limit uniformization is not.

This gives a domain-theoretic interpretation of the successor/limit distinction in transfinite jump iteration. At a successor stage one forms a relativized halting problem. At a limit stage one uniformly packages a cofinal family of prior stages. The former extends continuously to ideals; the latter detects completion of an infinite directed chain and therefore fails to commute with its supremum.

Combining these operations gives transfinite closure ordinals. One uniformization gate at ω yields closure ordinal $\omega \cdot 2$. Gates at $\omega, 2\omega, 3\omega, \ldots$ yield closure ordinal ω^2. Indexing gates by Kleene's system of ordinal notations and activating all of them at once carries the construction to closure ordinal ω_1^{CK}, with least fixed point the ideal of hyperarithmetic degrees.

The individual ingredients used here are classical: ideal completion, jump ideals, uniform upper bounds, and transfinite jump hierarchies all have established literatures [3, 11]. The point of the present construction is the semantic decomposition: non-uniform jump closure reaches a Scott-continuous fixed point, whereas uniformization of an infinite hierarchy is a discontinuous escape operation that restarts the diagonal process, all the way up to the boundary at which recursive ordinal notations themselves run out.

2 Turing degrees and ideal completion

2.1 Turing degrees

Let $\mathbf{D}_T$ denote the partially ordered set of Turing degrees. We write $\mathbf{a} \leq_T \mathbf{b}$ when every representative of $\mathbf{a}$ is Turing reducible to a representative of $\mathbf{b}$. The least degree is denoted $\mathbf{0}$. The join is

$$\mathbf{a} \vee \mathbf{b} = \deg_T(A \oplus B).$$

The Turing jump is the map

$$J : \mathbf{D}_T \to \mathbf{D}_T, \qquad J(\mathbf{a}) = \mathbf{a}'.$$

It is monotone,

$$\mathbf{a} \leq_T \mathbf{b} \implies \mathbf{a}' \leq_T \mathbf{b}',$$

and strictly progressive,

$$\mathbf{a} <_T \mathbf{a}'. \tag{1}$$

The strictness is the relativized undecidability of the halting problem. See, for example, [4, 9].

We write

$$\mathbf{a}^{(0)} = \mathbf{a}, \qquad \mathbf{a}^{(n+1)} = (\mathbf{a}^{(n)})'.$$

2.2 Turing ideals

Definition 2.1. A Turing ideal is a nonempty set $I \subseteq \mathbf{D}_T$ such that

1. if $\mathbf{b} \leq_T \mathbf{a} \in I$, then $\mathbf{b} \in I$;
2. if $\mathbf{a}, \mathbf{b} \in I$, then $\mathbf{a} \vee \mathbf{b} \in I$.

Equivalently, since $\mathbf{D}_T$ is an upper semilattice, Turing ideals are directed lower subsets of $\mathbf{D}_T$. Let $\mathrm{Idl}(\mathbf{D}_T)$ denote their poset under inclusion. For a degree $\mathbf{a}$, write

$$\downarrow \mathbf{a} = \{\mathbf{b} : \mathbf{b} \leq_T \mathbf{a}\}.$$

The map

$$\eta : \mathbf{D}_T \to \mathrm{Idl}(\mathbf{D}_T), \qquad \eta(\mathbf{a}) = \downarrow \mathbf{a}$$

embeds degrees as principal ideals.

The ideal completion is a dcpo. Directed suprema are unions:

$$\bigvee_{i \in D} I_i = \bigcup_{i \in D} I_i$$

for every directed family $(I_i)_{i \in D}$. In the usual domain-theoretic interpretation, the principal ideals are the finitely generated approximants and arbitrary ideals are their directed completions [1].

2.3 Canonical lifting to the ideal completion

We first record the general order-theoretic fact on which the construction rests.

Proposition 2.2 (Ideal-completion extension)**.** *Let P be a poset and let $f : P \to P$ be monotone. Define $\widehat{f} : \mathrm{Idl}(P) \to \mathrm{Idl}(P)$ by*

$$\widehat{f}(I) = \downarrow f[I] = \{y \in P : \exists x \in I \, (y \leq f(x))\}. \tag{2}$$

Then $\widehat{f}$ is well defined and Scott-continuous. Moreover,

$$\widehat{f}(\downarrow x) = \downarrow f(x) \tag{3}$$

for every $x \in P$.

Proof. The set $\widehat{f}(I)$ is lower by definition. Suppose $y_0 \leq f(x_0)$ and $y_1 \leq f(x_1)$ with $x_0, x_1 \in I$. Since I is directed, there is $x \in I$ with $x_0, x_1 \leq x$. Monotonicity gives $f(x_0), f(x_1) \leq f(x)$, so $f(x)$ is an upper bound in $\widehat{f}(I)$ for y_0, y_1. Thus $\widehat{f}(I)$ is an ideal.

Let $(I_i)_{i \in D}$ be directed. Then

$$\widehat{f}\Big(\bigcup_i I_i\Big) = \Big\{y : \exists x \in \bigcup_i I_i \, (y \leq f(x))\Big\} = \bigcup_i \{y : \exists x \in I_i \, (y \leq f(x))\} = \bigcup_i \widehat{f}(I_i).$$

Hence $\widehat{f}$ preserves directed suprema.

Finally, if $y \in \widehat{f}(\downarrow x)$, then $y \leq f(z)$ for some $z \leq x$, and hence $y \leq f(x)$. Conversely, $f(x)$ is among the generators, which proves (3). $\square$

The proposition exhibits the type shift that drives the paper: a progressive operation may fail to possess a fixed point among principal elements while its Scott-continuous extension possesses a nonprincipal fixed point in the completion.

3 The jump-closure fixed point

Apply Proposition 2.2 to the Turing jump.

Definition 3.1. Define $\Gamma : \mathrm{Idl}(\mathbf{D}_T) \longrightarrow \mathrm{Idl}(\mathbf{D}_T)$ by

$$\Gamma(I) = \downarrow \{\mathbf{a}' : \mathbf{a} \in I\}. \tag{4}$$

By Proposition 2.2, Γ is Scott-continuous and

$$\Gamma(\downarrow \mathbf{a}) = \downarrow \mathbf{a}'. \tag{5}$$

Since $\mathbf{a} \leq_T \mathbf{a}'$ for every degree $\mathbf{a}$, we also have

$$I \subseteq \Gamma(I). \tag{6}$$

Thus Γ is inflationary.

Proposition 3.2. *For a Turing ideal I, the following are equivalent:*

1. $\Gamma(I) = I$;
2. *I is closed under the Turing jump.*

Proof. If $\Gamma(I) = I$ and $\mathbf{a} \in I$, then $\mathbf{a}' \in \Gamma(I) = I$. Conversely, if I is jump-closed, then each generator $\mathbf{a}'$ in (4) already belongs to I, so downward closure gives $\Gamma(I) \subseteq I$. The reverse inclusion follows from (6). □

The fixed points of the lifted jump are therefore exactly the jump-closed Turing ideals.

For a degree $\mathbf{a}$, define

$$\mathrm{Arith}(\mathbf{a}) = \bigcup_{n<\omega} \downarrow \mathbf{a}^{(n)}. \tag{7}$$

Theorem 3.3 (Least jump fixed point)**.** *Starting from $\downarrow \mathbf{a}$, the finite iterates of Γ satisfy*

$$\Gamma^n(\downarrow \mathbf{a}) = \downarrow \mathbf{a}^{(n)} \tag{8}$$

for every $n < \omega$, and

$$\mathrm{lfp}_{\geq \downarrow \mathbf{a}}(\Gamma) = \bigcup_{n<\omega} \downarrow \mathbf{a}^{(n)} = \mathrm{Arith}(\mathbf{a}). \tag{9}$$

The closure ordinal of this iteration is exactly ω.

Proof. Equation (8) follows by induction from (5). Since Γ is Scott-continuous,

$$\Gamma\Big(\bigcup_{n<\omega} \downarrow \mathbf{a}^{(n)}\Big) = \bigcup_{n<\omega} \Gamma(\downarrow \mathbf{a}^{(n)}) = \bigcup_{n<\omega} \downarrow \mathbf{a}^{(n+1)} = \bigcup_{n<\omega} \downarrow \mathbf{a}^{(n)}.$$

Thus the union is fixed. It is least among fixed ideals containing $\downarrow \mathbf{a}$, because every such ideal must contain every finite jump $\mathbf{a}^{(n)}$ and hence everything below one of them.

No finite stage is fixed, since (1) gives $\mathbf{a}^{(n)} <_T \mathbf{a}^{(n+1)}$. Hence the first fixed stage is ω. □

For $\mathbf{a} = \mathbf{0}$, Post's theorem identifies (7) with the degrees of the arithmetical sets [6].

Corollary 3.4. *The global least fixed point of* Γ *is*

$$\mathrm{lfp}(\Gamma) = \mathrm{ARITH} = \bigcup_{n<\omega} \downarrow 0^{(n)},$$

where ARITH *denotes the Turing ideal of arithmetical degrees.*

This has the standard reverse-mathematical interpretation that the second-order part of an ω-model of ACA_0 is a Turing ideal closed under jump, while the arithmetical sets form the least such ω-model [10].

Corollary 3.5 (No principal fixed point)**.** *No principal Turing ideal is a fixed point of* Γ*.*

Proof. If $\Gamma(\downarrow \mathbf{a}) = \downarrow \mathbf{a}$, then by (5), $\downarrow \mathbf{a}' = \downarrow \mathbf{a}$, hence $\mathbf{a}' = \mathbf{a}$, contradicting (1). $\square$

Thus completion has not manufactured a degree that computes its own halting problem. The fixed point is intrinsically nonprincipal. It is better understood as a domain of solvers closed under producing the next relativized halting problem.

4 Uniformization at a limit

The fixed point ARITH does not contain $0^{(\omega)}$. This is the key to continuing the hierarchy. Each degree $0^{(n)}$ belongs to ARITH, but the single oracle

$$0^{(\omega)} = \bigoplus_{n<\omega} 0^{(n)} \tag{10}$$

is not arithmetical. The issue is uniformity. Membership of every finite stage in an ideal says $\forall n\, 0^{(n)} \in I$, but it does not provide one element of I from which the whole sequence can be uniformly recovered. Classical work on uniform upper bounds for Turing ideals studies a related and substantially more general notion [3]; here we use only the canonical uniform code supplied by a specified jump hierarchy.

Definition 4.1 (Uniformization gate)**.** Let $\mathbf{a}_0 <_T \mathbf{a}_1 <_T \mathbf{a}_2 <_T \cdots$ be a strictly increasing sequence of Turing degrees. Put

$$A = \bigcup_{n<\omega} \downarrow \mathbf{a}_n. \tag{11}$$

Let $\mathbf{b}$ satisfy

$$\mathbf{a}_n \leq_T \mathbf{b} \text{ for every } n, \tag{12}$$

and

$$\mathbf{b} \notin A. \tag{13}$$

Define

$$U_{A,\mathbf{b}}(I) = \begin{cases} \mathrm{Idl}(I \cup \{\mathbf{b}\}), & A \subseteq I, \\ I, & A \not\subseteq I. \end{cases} \tag{14}$$

The operator waits for completion of the entire chain and then makes the designated uniform upper bound available.

Theorem 4.2 (Uniformization is discontinuous)**.** *The operator* $U_{A,\mathbf{b}}$ *is monotone and inflationary, but it is not Scott-continuous.*

Proof. Inflationarity is immediate. For monotonicity, suppose $I \subseteq J$. If $A \not\subseteq I$, then either $A \not\subseteq J$, in which case $U(I) = I \subseteq J = U(J)$, or $A \subseteq J$, in which case $I \subseteq J \subseteq U(J)$. If $A \subseteq I$, then $A \subseteq J$, and ideal generation is monotone.

For discontinuity, take $I_n =\downarrow \mathbf{a}_n$. Then $(I_n)_{n<\omega}$ is an increasing chain and $\bigcup_n I_n = A$. For every n, $A \not\subseteq I_n$ because $\mathbf{a}_{n+1} \not\leq_T \mathbf{a}_n$. Hence $U(I_n) = I_n$ and

$$\bigcup_n U(I_n) = A. \tag{15}$$

On the other hand, $U(A) = \mathrm{Idl}(A \cup \{\mathbf{b}\})$. By (12), every member of A lies below $\mathbf{b}$, so $U(A) =\downarrow \mathbf{b}$. By (13), this strictly contains A. Therefore

$$U\Big(\bigcup_n I_n\Big) \neq \bigcup_n U(I_n),$$

so U is not Scott-continuous. □

The theorem identifies the exact informational source of the discontinuity: the condition that the entire chain has been completed cannot be witnessed at any finite member of the directed system.

4.1 The ω-jump gate

Take $\mathbf{a}_n = 0^{(n)}$ and $\mathbf{b} = 0^{(\omega)}$, where $0^{(\omega)}$ is given by (10). Let

$$A_\omega = \bigcup_{n<\omega} \downarrow 0^{(n)} = \mathrm{ARITH}.$$

Clearly $0^{(n)} \leq_T 0^{(\omega)}$ for every n. Moreover,

$$0^{(\omega)} \notin A_\omega. \tag{16}$$

Indeed, if $0^{(\omega)} \leq_T 0^{(n)}$ for some n, then since $0^{(n+1)} \leq_T 0^{(\omega)}$, we would have $0^{(n+1)} \leq_T 0^{(n)}$, contradicting strictness of the jump.

Let U_ω denote the corresponding instance of (14). Then

$$U_\omega(A_\omega) =\downarrow 0^{(\omega)}. \tag{17}$$

Thus A_ω, although fixed by jump closure, can be escaped by uniformizing the information distributed through it. Once this happens,

$$\Gamma(\downarrow 0^{(\omega)}) =\downarrow 0^{(\omega+1)},$$

and diagonalization resumes.

Schematically,

$$\text{jump iteration} \longrightarrow \text{jump-closed fixed ideal} \longrightarrow \text{uniform limit code} \longrightarrow \text{new jump iteration}. \tag{18}$$

The transfinite hierarchy is therefore not a sequence in which diagonalization is eventually defeated. Fixed closure and renewed diagonalization alternate.

5 A closure ordinal of $\omega \cdot 2$

We now combine the two operations into a single monotone transfinite induction. Define

$$\Theta_2 = U_\omega \circ \Gamma. \tag{19}$$

Since both factors are monotone and inflationary, so is Θ_2. By Theorem 4.2, it is not Scott-continuous.

Starting with $I_0 = \downarrow 0$, define

$$I_{\alpha+1} = \Theta_2(I_\alpha), \tag{20}$$

and, at limit ordinals,

$$I_\lambda = \bigcup_{\alpha<\lambda} I_\alpha. \tag{21}$$

For $n < \omega$, the gate has not fired, so

$$I_n = \downarrow 0^{(n)}. \tag{22}$$

Consequently,

$$I_\omega = A_\omega = \text{ARITH}. \tag{23}$$

At this point ordinary jump closure has stabilized, $\Gamma(I_\omega) = I_\omega$, but the uniformization gate fires:

$$I_{\omega+1} = \downarrow 0^{(\omega)}. \tag{24}$$

Successive stages then satisfy

$$I_{\omega+n+1} = \downarrow 0^{(\omega+n)} \qquad (n < \omega). \tag{25}$$

Put

$$A_{\omega\cdot 2} = \bigcup_{n<\omega} \downarrow 0^{(\omega+n)}. \tag{26}$$

(The earlier finite-jump ideals are already contained in this union.)

Theorem 5.1. *The closure ordinal of Θ_2 from $\downarrow 0$ is exactly $\omega \cdot 2$, and*

$$\text{lfp}(\Theta_2) = A_{\omega\cdot 2}.$$

Proof. Equations (22)–(25) determine the iteration below $\omega \cdot 2$. At the limit, $I_{\omega\cdot 2} = A_{\omega\cdot 2}$. This ideal is jump-closed. If $\mathbf{x} \leq_T 0^{(\omega+n)}$, then $\mathbf{x}' \leq_T 0^{(\omega+n+1)}$, which is again represented in the union. Hence $\Gamma(A_{\omega\cdot 2}) = A_{\omega\cdot 2}$. The U_ω gate adds nothing new because $0^{(\omega)} \in A_{\omega\cdot 2}$. Therefore $\Theta_2(A_{\omega\cdot 2}) = A_{\omega\cdot 2}$.

No earlier stage is fixed: every principal stage is moved by strictness of the jump, while A_ω is moved by the uniformization gate. Thus the first fixed stage is $\omega \cdot 2$. $\square$

The ordinal $\omega \cdot 2$ has a direct meaning in this semantics. The first ω-block closes under all finite jumps. Limit uniformization packages that block into $0^{(\omega)}$. A second ω-block then closes under every finite jump relative to this new oracle.

Notice that $A_{\omega\cdot 2}$ again contains every $0^{(\omega+n)}$ but does not contain the next uniform limit oracle $0^{(\omega\cdot 2)}$. The same mechanism can therefore be repeated.

6 Finite blocks and closure ordinal ω^2

We now iterate the construction through all finite multiples of ω. Since all ordinals below ω^2 have canonical notation $\omega k + n$ with $k, n < \omega$, no issues of notation invariance arise at this stage.

For $\alpha < \omega^2$, write $j_\alpha = 0^{(\alpha)}$. Successor stages satisfy

$$j_{\alpha+1} = j'_\alpha. \tag{27}$$

For $k \geq 1$, choose the standard limit representative

$$0^{(\omega k)} = \bigoplus_{n<\omega} 0^{(\omega(k-1)+n)}. \tag{28}$$

Because the displayed sequence is cofinal below ωk and higher finite jumps compute all lower stages, this degree uniformly codes the entire preceding initial segment.

For a nonzero limit $\lambda = \omega k < \omega^2$, put

$$A_\lambda = \bigcup_{\beta<\lambda} \downarrow j_\beta. \tag{29}$$

Define the corresponding gate

$$U_\lambda(I) = \begin{cases} \mathrm{Idl}(I \cup \{j_\lambda\}), & A_\lambda \subseteq I, \\ I, & A_\lambda \not\subseteq I. \end{cases} \tag{30}$$

Every U_λ is monotone and non-Scott-continuous by Theorem 4.2; a cofinal chain witnessing discontinuity is

$$\downarrow j_{\omega(k-1)} \subseteq \downarrow j_{\omega(k-1)+1} \subseteq \cdots .$$

For $m \geq 1$, define a combined uniformization operator

$$U_m(I) = \mathrm{Idl}\big(I \cup \{j_{\omega k} : 1 \leq k < m \text{ and } A_{\omega k} \subseteq I\}\big). \tag{31}$$

Thus U_m contains gates at $\omega, 2\omega, \ldots, \omega(m-1)$. Put

$$\Theta_m = U_m \circ \Gamma. \tag{32}$$

For $m = 1$, there are no gates and $\Theta_1 = \Gamma$.

Theorem 6.1 (Finite block theorem)**.** *For every integer $m \geq 1$, the closure ordinal of Θ_m, starting from $\downarrow 0$, is ωm, and its least fixed point is*

$$A_{\omega m} = \bigcup_{\beta<\omega m} \downarrow 0^{(\beta)}. \tag{33}$$

Proof. The first block is the ordinary jump iteration: $I_n = \downarrow j_n$ $(n < \omega)$, so $I_\omega = A_\omega$. If $m = 1$, there is no gate at ω, and Theorem 3.3 gives the result.

Assume $m > 1$. At A_ω, jump closure adds nothing while the gate U_ω adds j_ω. Hence $I_{\omega+1} = \downarrow j_\omega$. Repeated jump closure gives $I_{\omega+n+1} = \downarrow j_{\omega+n}$ $(n < \omega)$. At the next limit, $I_{\omega\cdot 2} = A_{\omega\cdot 2}$. If $2 < m$, the $U_{\omega\cdot 2}$ gate fires and yields $I_{\omega\cdot 2+1} = \downarrow j_{\omega\cdot 2}$. Continuing inductively, for every $1 \leq k < m$,

$$I_{\omega k} = A_{\omega k} \tag{34}$$

and

$$I_{\omega k+n+1} = \downarrow j_{\omega k+n} \qquad (n < \omega). \tag{35}$$

At stage ωm we obtain $A_{\omega m}$.

This ideal is jump-closed. If $\mathbf{x} \leq_T j_\beta$ for some $\beta < \omega m$, then $\mathbf{x}' \leq_T j_{\beta+1}$, and $\beta + 1 < \omega m$ because ωm is a limit ordinal. Thus $\Gamma(A_{\omega m}) = A_{\omega m}$.

Every gate appearing in U_m corresponds to some $\omega k < \omega m$, and its limit degree $j_{\omega k}$ already belongs to $A_{\omega m}$. Hence $U_m(A_{\omega m}) = A_{\omega m}$, so $A_{\omega m}$ is fixed.

No earlier stage is fixed. At a successor position strict jump progression forces movement. The only nonzero limit ordinals below ωm are the block boundaries ωk for $1 \leq k < m$, and at each of those the corresponding uniformization gate forces movement. Hence the closure ordinal is exactly ωm. □

We can activate all finite block gates simultaneously. Define

$$U_{<\omega^2}(I) = \mathrm{Idl}\big(I \cup \{j_{\omega k} : k \geq 1 \text{ and } A_{\omega k} \subseteq I\}\big) \tag{36}$$

and

$$\Theta_{<\omega^2} = U_{<\omega^2} \circ \Gamma. \tag{37}$$

Theorem 6.2 (ω^2-closure). *The closure ordinal of $\Theta_{<\omega^2}$ from $\downarrow 0$ is exactly ω^2, and its least fixed point is*

$$A_{\omega^2} = \bigcup_{\beta<\omega^2} \downarrow 0^{(\beta)}. \tag{38}$$

Proof. For every finite m, the iteration agrees below ωm with the operator Θ_m. Hence $I_{\omega m} = A_{\omega m}$. Taking the directed union over m gives

$$I_{\omega^2} = \bigcup_{m<\omega} A_{\omega m} = A_{\omega^2}.$$

The ideal A_{ω^2} is jump-closed because $\beta < \omega^2$ implies $\beta + 1 < \omega^2$. Thus $\Gamma(A_{\omega^2}) = A_{\omega^2}$. Every limit degree explicitly adjoined by $U_{<\omega^2}$ has the form $j_{\omega k}$ for finite k, hence already belongs to A_{ω^2}. Therefore $U_{<\omega^2}(A_{\omega^2}) = A_{\omega^2}$. So A_{ω^2} is fixed.

If $\alpha < \omega^2$, then either $\alpha = \omega k + n$ with $n > 0$, where successor jumping prevents stabilization, or $\alpha = \omega k$ with $k \geq 1$, where the corresponding uniformization gate prevents stabilization. Therefore no earlier stage is fixed. □

For every $m \geq 2$, and likewise for $\Theta_{<\omega^2}$, the operator is necessarily non-Scott-continuous. This follows abstractly because a Scott-continuous self-map of a pointed dcpo reaches its least fixed point at or before its ordinary ω-chain. More concretely, the chain $\downarrow 0 \subseteq \downarrow 0' \subseteq \cdots$ already witnesses the failure: the operator applied to the union can adjoin $0^{(\omega)}$, whereas no finite stage can.

7 Halting problems, transfinite jumps, and ordinal analysis

7.1 What is fixed?

There are three distinct objects that should not be conflated.

First, an individual degree $0^{(n)}, 0^{(\omega)}, \ldots$ is never fixed under the jump.

Second, an ideal such as $A_\omega = \bigcup_{n<\omega} \downarrow 0^{(n)}$ is fixed under non-uniform jump closure: $\Gamma(A_\omega) = A_\omega$.

Third, the uniform limit degree $0^{(\omega)}$ packages the preceding hierarchy into one oracle and therefore lies outside the fixed ideal it codes. The structure is

$$A_\omega \xmapsto{U_\omega} \downarrow 0^{(\omega)} \xmapsto{\Gamma} \downarrow 0^{(\omega+1)} \longmapsto \cdots .$$

A fixed ideal means that every individual member's halting problem is represented somewhere else inside the same domain. It does not mean that the domain has a single member solving all those problems uniformly. Once such a uniform member is introduced, it becomes subject to its own diagonal.

This gives the main conceptual principle: closure under relative halting can stabilize non-uniformly. Uniformizing the stabilized hierarchy creates a new oracle and therefore a new relativized halting problem.

7.2 Successor and limit stages

Classical transfinite jump iteration already distinguishes these operations. At a successor,

$$0^{(\alpha+1)} = (0^{(\alpha)})'. \tag{39}$$

At a recursive limit, an effective join codes a cofinal family of preceding stages,

$$0^{(\lambda)} \equiv_T \bigoplus_{n<\omega} 0^{(\lambda[n])}, \tag{40}$$

where $\lambda[0] < \lambda[1] < \cdots$ is an effective fundamental sequence cofinal in λ. The exact formulation is carried out using recursive ordinal notations; Spector's work gives the relevant invariance results for the resulting degrees [12, 7].

Equations (39) and (40) are computationally different, and the present semantics turns the difference into a continuity distinction. Successor jump closure extends to a Scott-continuous operation on ideals. Limit uniformization asks whether an entire cofinal family has become available. It therefore depends on infinitary information and fails Scott continuity.

In the language of inductive definitions, a limit gate has an infinitary premise. Aczel's analysis of monotone induction emphasizes precisely the relation between rule structure, transfinite stages, proof-tree ranks, and closure ordinals [2]. The present construction suggests the schematic correspondence

$$\text{successor jump} \leftrightarrow \text{continuous closure}, \qquad \text{limit jump} \leftrightarrow \text{discontinuous uniformization}.$$

7.3 Why ω^2 is not a degree invariant

Theorem 6.2 does not assert that $0^{(\omega)}$ has intrinsic computational complexity ω, that $0^{(\omega\cdot 2)}$ has intrinsic complexity $\omega \cdot 2$, or that a Turing degree determines a unique domain-theoretic closure ordinal.

Closure ordinals are invariants of operators and approximation schemes, not of Turing degrees alone. The ordinal ω^2 appears because the operator has been given a particular dependency architecture:

1. close under successor jumps through an ω-block;
2. wait for the whole block;
3. uniformly code it;

4. begin a new block;
5. repeat this process ω many times.

Different operational semantics can attach different closure ordinals to the same eventual degree-theoretic information. This is exactly where the connection to ordinal analysis becomes substantive: ordinals measure the architecture of the inductive process by which information becomes available.

The appropriate question is therefore not "What ordinal is the degree $\mathbf{a}$?" but "What closure ordinal is required by a specified rule system for generating or uniformly organizing degrees up to $\mathbf{a}$?" This is structurally analogous to the use of closure ordinals in theories of inductive definitions.

7.4 Toward the Church–Kleene boundary

The hierarchy above stops at ω^2 only to keep the mechanism transparent. Classical hyperarithmetic theory iterates the jump through every recursive ordinal below the Church–Kleene ordinal ω_1^{CK} [12, 7]. At each recursive limit λ, an effective fundamental sequence gives an increasing chain

$$\downarrow 0^{(\lambda[0])} \subseteq \downarrow 0^{(\lambda[1])} \subseteq \cdots .$$

Its union need not contain the uniform limit degree $0^{(\lambda)}$. The associated limit gate is therefore non-Scott-continuous by the same argument as Theorem 4.2.

This suggests a general program. Fix an effective ordinal notation system and equip recursive limit notations with uniformization rules. The resulting monotone operator on an appropriate domain of degree information will have a closure ordinal determined by the dependency structure of those limit rules.

Developing this systematically through arbitrary recursive ordinals requires care. Stage indices need not coincide naively with the ordinal labels on jump degrees: application of a discontinuous limit rule occurs only after the directed supremum at its triggering stage. Moreover, notation invariance must be derived from the classical machinery of hyperarithmetic hierarchies rather than assumed.

Problem 7.1. Construct a canonical family of monotone domain operators Θ_α for recursive ordinals α such that:

1. the fixed point of Θ_α contains exactly an intended initial segment of the transfinite jump hierarchy;
2. its closure rank is invariant under the chosen representation of α, up to an explicitly characterized ordinal transformation;
3. failures of Scott continuity occur precisely at effective limit-uniformization nodes.

A satisfactory solution would connect closure ordinals of domain operators directly to the structure of the hyperarithmetic jump hierarchy. At the Church–Kleene boundary, a further transition occurs: there is no recursive notation for ω_1^{CK} itself. Kleene's $\mathcal{O}$ and the hyperjump then encode a higher level of effective ordinal information. Whether this transition admits an analogous fixed-point/escape semantics is a natural continuation of the present framework. We give a full solution to Problem 7.1, including the transition at the Church–Kleene boundary, in Section 8 below.

7.5 Closure ordinals versus proof-theoretic ordinals

The closure ordinal of an inductive operator should not be identified with the proof-theoretic ordinal of a formal theory describing that operator. A closure ordinal measures how long a specified semantic induction takes to stabilize. A proof-theoretic ordinal measures the transfinite induction strength of a formal system. Classical theories of inductive definitions sharply distinguish these two roles [2].

The ordinal-analysis program suggested here therefore has two layers. First, classify operator schemes by semantic closure ordinals $\mathrm{cl}(\Theta)$. Second, ask for the proof-theoretic strength of theories capable of proving the existence, invariance, or well-foundedness of the corresponding fixed-point constructions. Sections 3–7 address the first layer only through ω^2; Section 8 carries it to ω_1^{CK}.

8 The Church–Kleene closure ordinal

Sections 5 and 6 stop at ω^2 so that the two mechanisms, continuous jump closure and discontinuous limit uniformization, can be watched separately. This section carries the same construction through every recursive ordinal at once. The result is a single monotone operator $\Theta_{\mathcal{O}}$ on $\mathrm{Idl}(\mathbf{D}_T)$ whose least fixed point is the ideal of hyperarithmetic degrees and whose closure ordinal is exactly ω_1^{CK}, together with truncations $\Theta_{<\lambda}$ that realize every recursive limit ordinal λ as a closure ordinal. This answers Problem 7.1. The only inputs beyond Sections 2–6 are Kleene's system of ordinal notations and Spector's invariance theorem for the H-sets.

8.1 Kleene's $\mathcal{O}$ and the H-sets

We recall the standard definitions; see Kleene [5], Spector [12], and Sacks [7, Ch. II].

Definition 8.1 (Kleene's $\mathcal{O}$)**.** The set $\mathcal{O} \subseteq \omega$ of *ordinal notations*, the relation $<_{\mathcal{O}}$, and the map $a \mapsto |a|$ are generated inductively by:

1. $1 \in \mathcal{O}$ and $|1| = 0$;
2. if $a \in \mathcal{O}$ then $2^a \in \mathcal{O}$, $|2^a| = |a| + 1$, and $b <_{\mathcal{O}} 2^a$ iff $b \leq_{\mathcal{O}} a$;
3. if φ_e is total, $\varphi_e(n) \in \mathcal{O}$ and $\varphi_e(n) <_{\mathcal{O}} \varphi_e(n+1)$ for every n, then $3 \cdot 5^e \in \mathcal{O}$, $|3 \cdot 5^e| = \sup_n |\varphi_e(n)|$, and $b <_{\mathcal{O}} 3 \cdot 5^e$ iff $b <_{\mathcal{O}} \varphi_e(n)$ for some n.

A notation of the form $3 \cdot 5^e$ is a *limit notation.* The set $\{|a| : a \in \mathcal{O}\}$ is exactly the set of recursive ordinals, i.e. the ordinals below ω_1^{CK}.

Definition 8.2 (H-sets)**.** For $a \in \mathcal{O}$ define $H_a \subseteq \omega$ by recursion along $<_{\mathcal{O}}$:

$$H_1 = \varnothing, \qquad H_{2^a} = (H_a)', \qquad H_{3 \cdot 5^e} = \{\langle n, x \rangle : x \in H_{\varphi_e(n)}\}.$$

Write $\mathbf{h}_a = \deg_T(H_a)$.

Thus the successor clause is a relativized halting problem and the limit clause is the effective join of the sets along the fundamental sequence coded by e. These are the operations (39) and (40) of Section 7.2, now indexed by notations rather than by ordinals.

Fact 8.3 (Kleene, Spector)**.**

(i) If $b <_{\mathcal{O}} a$ then $H_b \leq_T H_a$.

(ii) (Spector) *If* $|a| = |b|$ *then* $H_a \equiv_T H_b$.

(iii) For every $a \in \mathcal{O}$ *and every* $\beta < |a|$ *there is* $b <_{\mathcal{O}} a$ *with* $|b| = \beta$.

(iv) A set is hyperarithmetic (Δ^1_1) *iff it is computable from* H_a *for some* $a \in \mathcal{O}$.

By (ii) the degree $\mathbf{h}_a$ depends only on $|a|$, so the following notation is well defined.

Definition 8.4. For a recursive ordinal α let

$$\mathbf{0}^{(\alpha)} = \mathbf{h}_a \quad \text{for any } a \in \mathcal{O} \text{ with } |a| = \alpha.$$

For a limit ordinal $\lambda \leq \omega_1^{\mathrm{CK}}$ put

$$A_\lambda = \bigcup_{\beta<\lambda} \downarrow \mathbf{0}^{(\beta)}, \qquad \text{and} \qquad \mathrm{HYP} = A_{\omega_1^{\mathrm{CK}}} = \bigcup_{\alpha<\omega_1^{\mathrm{CK}}} \downarrow \mathbf{0}^{(\alpha)}.$$

For $\alpha < \omega^2$ this agrees, up to Turing degree, with the representatives fixed in (27)–(28): the canonical notations for $\omega k + n$ are built from the fundamental sequences $\omega(k-1)+n$, and the corresponding H-sets are exactly the joins displayed there. The ideals $A_\omega = \mathrm{ARITH}$, $A_{\omega\cdot 2}$, $A_{\omega m}$, A_{ω^2} of Sections 3–6 are the instances of A_λ for those λ.

Lemma 8.5.

(a) If $\beta < \gamma < \omega_1^{\mathrm{CK}}$ *then* $\mathbf{0}^{(\beta)} <_T \mathbf{0}^{(\gamma)}$*; more precisely* $\mathbf{0}^{(\beta+1)} \leq_T \mathbf{0}^{(\gamma)}$.

(b) $\mathbf{0}^{(\alpha+1)} = (\mathbf{0}^{(\alpha)})'$ *for every recursive* α.

(c) For every limit $\lambda \leq \omega_1^{\mathrm{CK}}$, A_λ *is a jump-closed Turing ideal. If* $\lambda < \omega_1^{\mathrm{CK}}$ *then* $\mathbf{0}^{(\lambda)} \notin A_\lambda$.

(d) HYP *is the Turing ideal of hyperarithmetic degrees.*

Proof. (b) is immediate from $H_{2^a} = (H_a)'$ and Fact 8.3(ii).

(a) Choose $c \in \mathcal{O}$ with $|c| = \gamma$. If $\beta + 1 < \gamma$, Fact 8.3(iii) gives $b <_{\mathcal{O}} c$ with $|b| = \beta + 1$, and then $\mathbf{0}^{(\beta+1)} = \mathbf{h}_b \leq_T \mathbf{h}_c = \mathbf{0}^{(\gamma)}$ by (i) and (ii). If $\beta + 1 = \gamma$ the inequality is an identity. Strictness $\mathbf{0}^{(\beta)} <_T \mathbf{0}^{(\beta+1)}$ is (1) together with (b).

(c) By (a) the family $\{\downarrow \mathbf{0}^{(\beta)} : \beta < \lambda\}$ is a chain, so A_λ is a directed lower set, i.e. a Turing ideal. If $\mathbf{x} \leq_T \mathbf{0}^{(\beta)}$ with $\beta < \lambda$ then $\mathbf{x}' \leq_T \mathbf{0}^{(\beta+1)}$ by (b), and $\beta + 1 < \lambda$ because λ is a limit; so A_λ is jump-closed. If $\lambda < \omega_1^{\mathrm{CK}}$ and $\mathbf{0}^{(\lambda)} \leq_T \mathbf{0}^{(\beta)}$ for some $\beta < \lambda$, then by (a) $\mathbf{0}^{(\beta+1)} \leq_T \mathbf{0}^{(\lambda)} \leq_T \mathbf{0}^{(\beta)}$, contradicting strictness of the jump.

(d) is (c) for $\lambda = \omega_1^{\mathrm{CK}}$ together with Fact 8.3(iv). □

8.2 The $\mathcal{O}$-gate

Every limit notation supplies an instance of Definition 4.1.

Lemma 8.6. *Let* $a = 3 \cdot 5^e$ *be a limit notation with* $|a| = \lambda$. *Put* $\mathbf{a}_n = \mathbf{h}_{\varphi_e(n)}$ *and* $\mathbf{b} = \mathbf{h}_a = \mathbf{0}^{(\lambda)}$. *Then* $(\mathbf{a}_n)_{n<\omega}$ *is strictly increasing,* $\bigcup_n \downarrow \mathbf{a}_n = A_\lambda$, *and* $\mathbf{b}$ *satisfies (12) and (13). Hence*

$$U_{A_\lambda, \mathbf{0}^{(\lambda)}}(I) = \begin{cases} \mathrm{Idl}(I \cup \{\mathbf{0}^{(\lambda)}\}), & A_\lambda \subseteq I, \\ I, & A_\lambda \not\subseteq I, \end{cases}$$

is monotone, inflationary, and not Scott-continuous.

Proof. Since $\varphi_e(n) <_{\mathcal{O}} \varphi_e(n+1)$ we have $|\varphi_e(n)| < |\varphi_e(n+1)|$, so the sequence is strictly increasing by Lemma 8.5(a). The inclusion $\bigcup_n \downarrow \mathbf{a}_n \subseteq A_\lambda$ holds because each $|\varphi_e(n)| < \lambda$; conversely if $\beta < \lambda = \sup_n |\varphi_e(n)|$ then $\beta < |\varphi_e(n)|$ for some n and $\mathbf{0}^{(\beta)} \leq_T \mathbf{a}_n$ by Lemma 8.5(a). Condition (12) is Fact 8.3(i), and (13) is Lemma 8.5(c). The last sentence is Theorem 4.2. □

The gate of Lemma 8.6 depends only on $\lambda = |a|$, not on the notation a or the fundamental sequence φ_e: its triggering condition is $A_\lambda \subseteq I$ and its output is the degree $\mathbf{0}^{(\lambda)}$. We therefore write it as U_λ. This is the representation invariance requested in Problem 7.1(2): it is inherited from Spector's theorem, exactly as anticipated in Section 7.4, and it is what allows all gates to be activated at once.

Definition 8.7 ($\mathcal{O}$-uniformization)**.** For a limit ordinal $\lambda \leq \omega_1^{\mathrm{CK}}$ define

$$U_{<\lambda}(I) = \mathrm{Idl}\big(I \cup \{\mathbf{0}^{(\mu)} : \mu < \lambda \text{ a limit ordinal and } A_\mu \subseteq I\}\big), \qquad \Theta_{<\lambda} = U_{<\lambda} \circ \Gamma.$$

In the extreme case write $U_{\mathcal{O}} = U_{<\omega_1^{\mathrm{CK}}}$ and $\Theta_{\mathcal{O}} = U_{\mathcal{O}} \circ \Gamma$.

Thus $\Theta_{<\omega} = \Gamma$, $\Theta_{<\omega\cdot 2} = \Theta_2$ of (19), $\Theta_{<\omega m} = \Theta_m$ of (32), and $\Theta_{<\omega^2} = \Theta_{<\omega^2}$ of (37). Definition 8.7 is (36) with the finite-multiple gates replaced by gates at every recursive limit below λ.

Proposition 8.8. *For every limit $\lambda \leq \omega_1^{\mathrm{CK}}$, $U_{<\lambda}$ and $\Theta_{<\lambda}$ are monotone and inflationary. If $\lambda > \omega$ they are not Scott-continuous.*

Proof. Inflationarity is immediate. If $I \subseteq J$ then every gate condition $A_\mu \subseteq I$ implies $A_\mu \subseteq J$, so the generating set for $U_{<\lambda}(I)$ is contained in that for $U_{<\lambda}(J)$, and ideal generation is monotone. Composition preserves both properties. For $\lambda > \omega$ the gate U_ω is present, and the chain $\downarrow \mathbf{0}^{(n)}$ witnesses discontinuity exactly as at the end of Section 6. □

Two facts about where gates fire will carry the induction. Both say that no gate ever fires early.

Lemma 8.9. *Let $\lambda \leq \omega_1^{\mathrm{CK}}$ be a limit ordinal.*

(a) For every recursive α, $U_{<\lambda}(\downarrow \mathbf{0}^{(\alpha)}) = \downarrow \mathbf{0}^{(\alpha)}$.

(b) For every limit $\mu < \lambda$, $U_{<\lambda}(A_\mu) = \downarrow \mathbf{0}^{(\mu)}$.

(c) $U_{<\lambda}(A_\lambda) = A_\lambda$ and $\Gamma(A_\lambda) = A_\lambda$; hence $\Theta_{<\lambda}(A_\lambda) = A_\lambda$.

Proof. (a) Suppose a gate at a limit $\nu < \lambda$ fires at $\downarrow \mathbf{0}^{(\alpha)}$, i.e. $A_\nu \subseteq \downarrow \mathbf{0}^{(\alpha)}$. If $\nu > \alpha$ then $\alpha + 1 < \nu$ and $\mathbf{0}^{(\alpha+1)} \in A_\nu$, so $\mathbf{0}^{(\alpha+1)} \leq_T \mathbf{0}^{(\alpha)}$, contradicting Lemma 8.5(a). Hence $\nu \leq \alpha$, and the adjoined degree $\mathbf{0}^{(\nu)}$ already lies in $\downarrow \mathbf{0}^{(\alpha)}$.

(b) Suppose a gate at a limit $\nu < \lambda$ fires at A_μ. If $\nu > \mu$ then $\mathbf{0}^{(\mu)} \in A_\nu \subseteq A_\mu$, contradicting Lemma 8.5(c). If $\nu < \mu$ then $\mathbf{0}^{(\nu)} \in A_\mu$ already. The gate at $\nu = \mu$ does fire, since $A_\mu \subseteq A_\mu$, and adjoins $\mathbf{0}^{(\mu)}$. Therefore $U_{<\lambda}(A_\mu) = \mathrm{Idl}(A_\mu \cup \{\mathbf{0}^{(\mu)}\}) = \downarrow \mathbf{0}^{(\mu)}$, the last step because $A_\mu \subseteq \downarrow \mathbf{0}^{(\mu)}$ by Lemma 8.5(a).

(c) Every gate in $U_{<\lambda}$ sits at some $\nu < \lambda$ and adjoins $\mathbf{0}^{(\nu)} \in A_\lambda$. Jump closure of A_λ is Lemma 8.5(c). □

8.3 Closure ordinal ω_1^{CK}

Theorem 8.10 (Church–Kleene closure)**.** *Let $\lambda \leq \omega_1^{\mathrm{CK}}$ be a limit ordinal, and iterate $\Theta_{<\lambda}$ from $I_0 = \downarrow \mathbf{0}$ by $I_{\alpha+1} = \Theta_{<\lambda}(I_\alpha)$ and $I_\nu = \bigcup_{\alpha<\nu} I_\alpha$ at limits. Then for every $\alpha \leq \lambda$:*

$$I_n = \downarrow \mathbf{0}^{(n)} \qquad (n < \omega), \tag{1}$$

$$I_\mu = A_\mu \qquad (\mu \leq \lambda \text{ a limit}), \tag{2}$$

$$I_{\mu+n+1} = \downarrow \mathbf{0}^{(\mu+n)} \qquad (\mu < \lambda \text{ a limit},\ n < \omega). \tag{3}$$

Consequently the closure ordinal of $\Theta_{<\lambda}$ from $\downarrow \mathbf{0}$ is exactly λ, and

$$\mathrm{lfp}(\Theta_{<\lambda}) = A_\lambda.$$

In particular $\Theta_{\mathcal{O}}$ has closure ordinal exactly ω_1^{CK} and

$$\mathrm{lfp}(\Theta_{\mathcal{O}}) = \mathrm{HYP}.$$

Proof. We prove (1)–(3) by induction on $\alpha \leq \lambda$. Two consequences of the three equations are used at limit steps: for every $\alpha < \lambda$,

$$\downarrow \mathbf{0}^{(\alpha)} \subseteq I_{\alpha+1} \qquad \text{and} \qquad I_\alpha \subseteq \downarrow \mathbf{0}^{(\alpha)}. \tag{4}$$

Indeed, reading α in the form n, μ, or $\mu+n$: $\downarrow \mathbf{0}^{(n)} \subseteq \downarrow \mathbf{0}^{(n+1)} = I_{n+1}$; $\downarrow \mathbf{0}^{(\mu)} = I_{\mu+1}$; $\downarrow \mathbf{0}^{(\mu+n)} = I_{\mu+n+1}$; and $I_n = \downarrow \mathbf{0}^{(n)}$, $I_\mu = A_\mu \subseteq \downarrow \mathbf{0}^{(\mu)}$, $I_{\mu+n+1} = \downarrow \mathbf{0}^{(\mu+n)} \subseteq \downarrow \mathbf{0}^{(\mu+n+1)}$, using Lemma 8.5(a) throughout.

Finite stages. $I_0 = \downarrow \mathbf{0}$. If $I_n = \downarrow \mathbf{0}^{(n)}$ then $\Gamma(I_n) = \downarrow \mathbf{0}^{(n+1)}$ by (5), and $U_{<\lambda}$ leaves this fixed by Lemma 8.9(a). This is (1).

Limit stages. Let $\mu \leq \lambda$ be a limit and assume the equations below μ. By (4),

$$A_\mu = \bigcup_{\alpha<\mu} \downarrow \mathbf{0}^{(\alpha)} \subseteq \bigcup_{\alpha<\mu} I_{\alpha+1} = I_\mu = \bigcup_{\alpha<\mu} I_\alpha \subseteq \bigcup_{\alpha<\mu} \downarrow \mathbf{0}^{(\alpha)} = A_\mu,$$

which is (2).

Successors of limits. Let $\mu < \lambda$ be a limit with $I_\mu = A_\mu$. Then $\Gamma(A_\mu) = A_\mu$ by Lemma 8.5(c), and $U_{<\lambda}(A_\mu) = \downarrow \mathbf{0}^{(\mu)}$ by Lemma 8.9(b); so $I_{\mu+1} = \downarrow \mathbf{0}^{(\mu)}$. If $I_{\mu+n+1} = \downarrow \mathbf{0}^{(\mu+n)}$, then Γ gives $\downarrow \mathbf{0}^{(\mu+n+1)}$ by (5) and Lemma 8.5(b), and $U_{<\lambda}$ leaves it fixed by Lemma 8.9(a). This is (3).

Fixed point at λ. By (2), $I_\lambda = A_\lambda$, and $\Theta_{<\lambda}(A_\lambda) = A_\lambda$ by Lemma 8.9(c).

No earlier fixed stage. Every $\alpha < \lambda$ has one of the forms n, μ, $\mu+n$. At $\alpha = n$ and $\alpha = \mu+n$, $I_{\alpha+1}$ is the principal ideal of the jump of the generator of I_α, so $I_\alpha \subsetneq I_{\alpha+1}$ by (1). At $\alpha = \mu$, $I_\mu = A_\mu \subsetneq \downarrow \mathbf{0}^{(\mu)} = I_{\mu+1}$ by Lemma 8.5(c).

Leastness. $\mathrm{Idl}(\mathbf{D}_T)$ is a complete lattice (intersections of Turing ideals are Turing ideals), so by the Knaster–Tarski theorem $\Theta_{<\lambda}$ has a least fixed point, and the transfinite iteration of a monotone map from the least element stabilizes exactly at it. Hence $\mathrm{lfp}(\Theta_{<\lambda}) = I_\lambda = A_\lambda$.

The final clause is the case $\lambda = \omega_1^{\mathrm{CK}}$, where $A_\lambda = \mathrm{HYP}$ by Lemma 8.5(d). $\square$

Theorems 3.3, 5.1, 6.1, and 6.2 are the cases $\lambda = \omega,\ \omega \cdot 2,\ \omega m,\ \omega^2$.

Corollary 8.11. *The ordinal at which $\mathbf{0}^{(\alpha)}$ first enters the iteration of $\Theta_{<\lambda}$ is*

$$s(\alpha) = \begin{cases} \alpha, & \alpha < \omega, \\ \alpha + 1, & \omega \leq \alpha < \lambda. \end{cases}$$

Proof. Read off (1)–(3): $\mathbf{0}^{(\mu+n)}$ first appears at stage $\mu+n+1$, which as an ordinal is $(\mu+n)+1$. $\square$

This is the "explicitly characterized ordinal transformation" of Problem 7.1(2). The lag is exactly one and does not accumulate, because a gate fires only after the directed supremum that completes its chain, and a single successor step then absorbs the lag. The stage index and the jump index agree again at every subsequent limit, where both equal the ordinal μ and the stage content is A_μ rather than a principal ideal.

8.4 Problem 7.1

Theorem 8.10 and its corollary answer the three parts of Problem 7.1 for the family $\{\Theta_{<\lambda} : \lambda \leq \omega_1^{\mathrm{CK}} \text{ a limit}\}$.

1. The least fixed point of $\Theta_{<\lambda}$ is exactly the initial segment A_λ of the transfinite jump hierarchy, and no ideal strictly between A_λ and any $\downarrow \mathbf{0}^{(\alpha)}$, $\alpha < \lambda$, is ever visited except the A_μ for limits $\mu < \lambda$.

2. The operator $\Theta_{<\lambda}$ is defined in terms of the ordinals $\mu < \lambda$ and the degrees $\mathbf{0}^{(\mu)}$, both of which are notation-independent by Spector's theorem. A construction that instead fixes, for each limit μ, a single notation a_μ and the gate of Lemma 8.6 for that notation yields literally the same operator, since the gate depends only on $|a_\mu|$. The stage/label transformation is Corollary 8.11.

3. Γ is Scott-continuous by Proposition 2.2, and every gate U_μ is non-Scott-continuous by Lemma 8.6, with the discontinuity witnessed by the chain along any fundamental sequence for μ. Since $\Theta_{<\lambda} = U_{<\lambda} \circ \Gamma$ and $U_{<\lambda}$ is the simultaneous activation of the gates U_μ, $\mu < \lambda$, the failures of Scott continuity are located precisely at the recursive limit ordinals below λ, which are the effective limit-uniformization nodes.

The construction also sharpens the reading of closure ordinals proposed in Section 7.3. The ordinal ω_1^{CK} is not extracted from the lattice by the iteration; it is inherited from the indexing of the gates by $\mathcal{O}$. What the iteration adds is the identification of that ordinal as the exact closure rank of a specific monotone operator, and the location of every one of its discontinuities.

8.5 At the boundary

At $\lambda = \omega_1^{\mathrm{CK}}$ the pattern of (18) stops for a purely notational reason: there is no limit notation with $|a| = \omega_1^{\mathrm{CK}}$, so $U_{\mathcal{O}}$ has no gate whose triggering chain is HYP. The ideal HYP is fixed in the sense of Section 7.1: every member's halting problem is represented inside it, non-uniformly, but no member codes the whole hierarchy.

The uniform code does exist one type level up. Let $\mathbf{o} = \deg_T(\mathcal{O})$. The set $\{\langle a, x\rangle : a \in \mathcal{O},\ x \in H_a\}$ is Turing-equivalent to $\mathcal{O}$, so $\mathbf{o}$ is a uniform upper bound for HYP, and $\mathbf{o} \notin \mathrm{HYP}$ since $\mathcal{O}$ is Π^1_1-complete and not Δ^1_1. Because HYP is a countable ideal, it is the union of some strictly increasing ω-chain of principal ideals, and Definition 4.1 applies: the *hyperjump gate*

$$U_{\mathrm{HYP},\mathbf{o}}(I) = \begin{cases} \mathrm{Idl}(I \cup \{\mathbf{o}\}), & \mathrm{HYP} \subseteq I, \\ I, & \text{otherwise}, \end{cases}$$

is monotone, inflationary, and non-Scott-continuous. The difference from the gates U_μ is that no witnessing chain is effective: there is no computable fundamental sequence for ω_1^{CK}. Composing

$U_{\mathrm{HYP},\mathbf{o}}$ with $\Theta_{\mathcal{O}}$ escapes HYP to $\downarrow \mathbf{o}$, after which $\Gamma(\downarrow \mathbf{o}) = \downarrow \mathbf{o}'$ and the diagonal resumes. Carrying the construction further requires relativizing $\mathcal{O}$ to $\mathbf{o}$, whose least admissible $\omega_1^{\mathcal{O}}$ is strictly larger than ω_1^{CK}; the closure ordinal then moves with the oracle. This is the fixed-point/escape semantics for the hyperjump asked about at the end of Section 7.4, and it is the natural next step of the framework.

9 Relation to bounded informational fixed points

A simpler domain-theoretic halting construction begins with finite observations of a single computation and advances them one step at a time. If p_n records the behavior of a machine through the first n stages, a local extension operator can produce an ascending chain

$$p_0 \sqsubseteq p_1 \sqsubseteq p_2 \sqsubseteq \cdots$$

whose Scott supremum is reached at ω. The important feature of such an operator is not a purported lower bound on literal self-simulation time, but its finite dependency structure: each newly defined coordinate depends on finite information from the previous approximation.

The jump-ideal construction is the degree-theoretic analogue of this mechanism. The operator Γ is Scott-continuous, so its iterative closure is exhausted by an ω-chain. The uniformization gates introduced above mark the first point at which the dependency structure ceases to be finite: a gate requires the entire cofinal chain before it can fire. Consequently, closure ordinal and continuity change together.

This comparison suggests a more general hierarchy of informational operators:

$$\text{finite witness dependence} \implies \text{Scott continuity and closure by } \omega,$$

$$\text{cofinal-limit dependence} \implies \text{non-Scott-continuity and transfinite closure,}$$

$$\text{nested effective limit dependence} \implies \text{higher recursive closure ranks.}$$

The Turing jump supplies a canonical family of computationally meaningful successor operations with which to populate such a hierarchy. The $\mathcal{O}$-indexed gates of Section 8 populate it all the way to ω_1^{CK}, and the hyperjump gate of Section 8.5 marks where a further, non-effective, layer would have to begin.

10 Discussion

The usual statement that the Turing jump has no fixed point is correct but incomplete as a description of iterated halting phenomena. It applies to individual degrees. Passing to the ideal completion changes the fixed-point structure without changing the underlying impossibility theorem.

The canonical ideal extension $\Gamma(I) = \downarrow \{\mathbf{a}' : \mathbf{a} \in I\}$ is Scott-continuous. Its least fixed point above $\mathbf{a}$ is $\bigcup_{n<\omega} \downarrow \mathbf{a}^{(n)}$. For $\mathbf{a} = \mathbf{0}$, this is the ideal of arithmetical degrees. There is therefore a precise sense in which the finite jump hierarchy converges to a fixed point. It is not the false equation

$$0^{(\omega)} = (0^{(\omega)})'.$$

It is the true equation

$$\Gamma(\mathrm{ARITH}) = \mathrm{ARITH}.$$

The contrast with the uniform limit oracle is fundamental:

$$0^{(\omega)} \notin \mathrm{ARITH}.$$

The first equation says that the collection is closed under taking relativized halting problems one member at a time. The second says that the collection does not uniformly internalize its entire construction in a single oracle.

Limit uniformization converts the former kind of closure into the latter kind of object. Because doing so requires recognition of completion of an infinite directed chain, it is not Scott-continuous. The new oracle is then subject to the jump again.

The recurrence of the diagonal can thus be pictured as

$$\text{successor jumps} \longrightarrow \text{fixed ideal} \longrightarrow \text{uniform limit oracle} \longrightarrow \text{new relative halting problem.}$$

Diagonalization is not eliminated at the limit but relocated. Non-uniform closure can absorb every previous jump operation; uniformly coding that closure creates a new point from which a further diagonal can be taken.

Theorems 3.3, 5.1, 6.2, and 8.10 exhibit the closure ordinals of this semantics:

$$\omega, \quad \omega \cdot 2, \quad \omega^2, \quad \dots, \quad \lambda, \quad \dots, \quad \omega_1^{\mathrm{CK}}.$$

Section 8 shows that this list is in fact exhaustive at the effective level: every recursive limit ordinal is the closure ordinal of some instance $\Theta_{<\lambda}$ of the construction, the whole family is realized by a single notation-independent operator $\Theta_{\mathcal{O}}$ with closure ordinal exactly ω_1^{CK} and least fixed point HYP, and the hyperjump gate of Section 8.5 shows how the same fixed-point/escape semantics continues one step beyond the boundary, at the cost of losing effectivity of the triggering chain.

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